Trạng thái	Đã xong
Bắt đầu vào lúc	Thứ Sáu, 23 tháng 8 2024, 1:55 PM
Kết thúc lúc	Thứ Sáu, 23 tháng 8 2024, 1:57 PM
Thời gian thực hiện	1 phút 33 giây
Điểm	5,00/5,00
Điểm	**10,00** trên 10,00 (**100**%)

Câu hỏi **1**

Đúng

Đạt điểm 1,00 trên 1,00

In the coordinate plane, we have class Point to store a point with it's x-y coordinate.

Your task in this exercise is to implement functions marked with /* * STUDENT ANSWER */.

Note: For exercises in Week 1, we have #include <bits/stdc++.h> and using namespace std;

For example:

Test	Result
`Point A(2, 3);` `cout << A.getX() << " " << A.getY();`	2 3

`Point A(2, 3);`
`Point B(1, 1);`
`cout << pow(A.distanceToPoint(B), 2);`
	5

Answer: (penalty regime: 0 %)

Reset answer

```
class Point
{
private:
    double x, y;

public:
    Point()
    {
        /*
         * STUDENT ANSWER
         * TODO: set zero x-y coordinate
         */
         x=y=0;
    }

    Point(double x, double y)
    {
        /*
         * STUDENT ANSWER
         */
         this->x=x;
         this->y=y;
    }

    void setX(double x)
    {
        /*
         * STUDENT ANSWER
         */
         this->x =x;
    }

    void setY(double y)
    {
        /*
         * STUDENT ANSWER
         */
           this->y=y;
    }

    double getX() const
    {
        /*
         * STUDENT ANSWER
```

```
        */
        return this->x;
    }

    double getY() const
    {
       /*
        * STUDENT ANSWER
        */
                return this->y;
    }

    double distanceToPoint(const Point& pointA)
    {
       /*
        * STUDENT ANSWER
        * TODO: calculate the distance from this point to point A in the coordinate plane
        */
        int x = pointA.x;
        int y= pointA.y;
        double distance= sqrt(pow((x-this->x),2) +  pow((y-this->y),2));
        return distance;

    }
};
```

	Test	Expected	Got	
✓	`Point A(2, 3);` `cout << A.getX() << " " << A.getY();`	2 3	2 3	✓

✓	`Point A(2, 3);`
	`Point B(1, 1);`
	`cout << pow(A.distanceToPoint(B), 2);`
		5	5	✓

Passed all tests! ✓

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi **2**

Đúng

Đạt điểm 1,00 trên 1,00

In the coordinate plane, a circle is defined by center and radius.

Your task in this exercise is to implement functions marked with /* * STUDENT ANSWER */.

Note: you can use implemented class Point in *previous question*

For example:

Test	Result
`Circle A;` `A.printCircle();`	`Center: {0.00, 0.00} and Radius 0.00`

Answer: (penalty regime: 0 %)

Reset answer

```
#include<iostream>
#include<string>
#include<cmath>


class Point
{
private:
    double x, y;

public:
    Point()
    {
        /*
         * STUDENT ANSWER
         * TODO: set zero x-y coordinate
         */
         x=y=0;
    }

    Point(double x, double y)
    {
        /*
         * STUDENT ANSWER
         */
         this->x=x;
         this->y=y;
    }

    void setX(double x)
    {
        /*
         * STUDENT ANSWER
         */
         this->x =x;
    }

    void setY(double y)
    {
        /*
         * STUDENT ANSWER
         */
           this->y=y;
    }
```

```
    double getX() const
    {
        /*
         * STUDENT ANSWER
         */
         return this->x;
    }

    double getY() const
    {
        /*
         * STUDENT ANSWER
         */
                 return this->y;
    }

    double distanceToPoint(const Point& pointA)
    {
        /*
         * STUDENT ANSWER
         * TODO: calculate the distance from this point to point A in the coordinate plane
         */
         int x = pointA.x;
         int y= pointA.y;
         double distance= sqrt(pow((x-this->x),2) +  pow((y-this->y),2));
         return distance;

    }
};

class Circle
{
private:
    Point center;
    double radius;

public:
    Circle()
    {
        /*
         * STUDENT ANSWER
         * TODO: set zero center's x-y and radius
         */
         center.setX(0);
         center.setY(0);
         radius=0;
    }

    Circle(Point c, double radius)
    {
        /*
         * STUDENT ANSWER
         */
         double x=c.getX();
         double y=c.getY();

        center.setX(x);
        center.setY(y);

         this->radius=radius;
    }

    Circle(const Circle &circle)
    {
        /*
         * STUDENT ANSWER
         */

          double x=circle.center.getX();
         double y=circle.center.getY();

         double radius= circle.getRadius();
```

```
        this->center.setX(x);
        this->center.setY(y);

        this->radius=radius;

    }

    void setCenter(Point point)
    {

        double x=point.getX();
        double y=point.getY();

        this->center.setX(x);
        this->center.setY(y);
    }

    void setRadius(double radius)
    {
        /*
         * STUDENT ANSWER
         */
        this->radius=radius;
    }

    Point getCenter() const
    {
        /*
         * STUDENT ANSWER
         */
        return this->center;
    }

    double getRadius() const
    {
        /*
         * STUDENT ANSWER
         */
        return radius;
    }

    void printCircle()
    {
        printf("Center: {%.2f, %.2f} and Radius %.2f\n", this->center.getX(), this->center.getY(), this->rad
    }
};
```

	Test	Expected	Got	
✓	Circle A; A.printCircle();	Center: {0.00, 0.00} and Radius 0.00	Center: {0.00, 0.00} and Radius 0.00	✓

Passed all tests! ✓

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi **3**

Đúng

Đạt điểm 1,00 trên 1,00

In a game, we have class `Character` to store characters' data.
The class `Character` is declared as below:

```
class Character {
protected:
    int hp;
    int x;
    int y;
public:
    // Constructor: set the values of x and y and hp to 0
    Character();

    // Constructor: set the values of hp, x and y to each parameter
    Character(int hp, int x, int y);

    // Set and get hp
    int getHp();
    void setHp(int hp);

    // Set and get x
    int getX();
    void setX(int x);

    // Set and get y
    int getY();
    void setY(int y);

    // Get Manhattan distance to other character
    int getManhattanDistTo(Character* other);
};
```

Your task is to define the constructors and the methods of the class.

Note:

In this task, `iostream` library has been included, and `namespace std` is being used. No other libraries are allowed.

For example:

Test	Result
`Character ch1(100, 3, 6);` `cout << ch1.getHp() << " " << ch1.getX() << " " << ch1.getY();`	`100 3 6`

Answer: (penalty regime: 0 %)

Reset answer

```
#include<cmath>

Character::Character() {
    // STUDENT ANSWER
    x=y=hp=0;
}

Character::Character(int hp, int x, int y) {
```

```cpp
Character::Character(int hp, int x, int y) {
    // STUDENT ANSWER
    this->hp=hp;
    this->x=x;
    this->y=y;
}

int Character::getHp() {
    // STUDENT ANSWER
    return hp;
}

void Character::setHp(int hp) {
    // STUDENT ANSWER
    this->hp=hp;
}

int Character::getX() {
    // STUDENT ANSWER
    return this->x;
}

void Character::setX(int x) {
    // STUDENT ANSWER
     this->x=x;
}

int Character::getY() {
    // STUDENT ANSWER
   return this->y;
}

void Character::setY(int y) {
    // STUDENT ANSWER
    this->y=y;
}

int Character::getManhattanDistTo(Character* other) {
    // STUDENT ANSWER
    int a,b;
    a= other->getX();
    b=other->getY();

     int distance= abs(a-this->x) +  abs(b-this->y);
     return distance;
}
```

	Test	Expected	Got	
✓	`Character ch1(100, 3, 6);` `cout << ch1.getHp() << " " << ch1.getX() << " " << ch1.getY();`	100 3 6	100 3 6	✓
✓	`Character ch2;` `cout << ch2.getHp() << " " << ch2.getX() << " " << ch2.getY();`	0 0 0	0 0 0	✓

✓	`Character* ch31 = new Character(100, 1, 2);`
	`Character* ch32 = new Character(100, -3, 4);`
	`cout << ch31->getManhattanDistTo(ch32);`
	`delete ch31;`
	`delete ch32;`
		6	6	✓

✓	`Character ch4;`
	`ch4.setX(4);`
	`cout << ch4.getX();`
		4	4	✓

	Test	Expected	Got	

✓	`Character ch5;`
	`ch5.setY(5);`
	`cout << ch5.getY();`
		5	5	✓

✓	`Character ch6;`
	`ch6.setHp(6);`
	`cout << ch6.getHp();`
		6	6	✓

Passed all tests! ✓

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi **4**

Đúng

Đạt điểm 1,00 trên 1,00

Hoang is a K19 student studying at Bach Khoa University. He plans to write a book management software for the library. In the class design, Hoang has designed the class Book as follows:

```
class Book
{
private:
    char* title;
    char* authors;
    int publishingYear;
public:
   // some method
}
```

Your task in this exercise is to implement functions marked with /* * STUDENT ANSWER   */.

Note: For exercises in Week 2, we have #include <bits/stdc++.h> and using namespace std;

For example:

Test	Result

`Book book1("Giai tich 1","Nguyen Dinh Huy",2000);`
`book1.printBook();`
	`Giai tich 1`
	`Nguyen Dinh Huy`
	`2000`

`Book book1("Giai tich 1","Nguyen Dinh Huy",2000);`
`Book book2 = book1;`
`book2.printBook();`
	`Giai tich 1`
	`Nguyen Dinh Huy`
	`2000`

Answer: (penalty regime: 0 %)

Reset answer

```
#include<iostream>
#include<cstring>

using namespace std;



class Book
{
private:
    char* title;
    char* authors;
    int publishingYear;

public:
    Book()
    {
        /*
         * STUDENT ANSWER
         * TODO: set zero publishingYear and null pointer
         */
         title=authors=nullptr;
         publishingYear=0;
    }

    Book(const char* title, const char* authors, int publishingYear)
```

```
    Book(const char* title, const char* authors, int publishingYear)
    {
        /*
         * STUDENT ANSWER
         */


         this->title= strdup(title);
         this->authors= strdup(authors);
         this->publishingYear=publishingYear;

    }

    Book(const Book &book)
    {
        /*
         * STUDENT ANSWER
         * TODO: deep copy constructor
         */
         char*a= book.getTitle();
         char*b= book.getAuthors();
         int c = book.getPublishingYear();

         this-> title =a;
         this->authors=b;
         this->publishingYear=c;

    }

    void setTitle(const char* title)
    {

         this->title= strdup(title);


    }

    void setAuthors(const char* authors)
    {
        /*
         * STUDENT ANSWER
         */
this->authors= strdup(authors);    }

    void setPublishingYear(int publishingYear)
    {
        /*
         * STUDENT ANSWER
         */
                  this->publishingYear=publishingYear;

    }

    char* getTitle() const
    {
        /*
         * STUDENT ANSWER
         */
         return this->title;
    }

    char* getAuthors() const
    {
        /*
         * STUDENT ANSWER
         */
         return this->authors;

    }

    int getPublishingYear() const
    {
        /*
```

```
        /*
         * STUDENT ANSWER
         */
        return this->publishingYear;
    }

    ~Book()
    {
        /*
         * STUDENT ANSWER
         */
        title=authors=nullptr;
        publishingYear=0;
    }

    void printBook(){
        printf("%s\n%s\n%d", this->title, this->authors, this->publishingYear);
    }
};
```

	Test	Expected	Got	
✓	Book book1("Giai tich 1","Nguyen Dinh Huy",2000); book1.printBook();	Giai tich 1 Nguyen Dinh Huy 2000	Giai tich 1 Nguyen Dinh Huy 2000	✓
✓	Book book1("Giai tich 1","Nguyen Dinh Huy",2000); Book book2 = book1; book2.printBook();	Giai tich 1 Nguyen Dinh Huy 2000	Giai tich 1 Nguyen Dinh Huy 2000	✓

Passed all tests! ✓

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi **5**

Đúng

Đạt điểm 1,00 trên 1,00

1. In the toy store, all toy has a price. Car toy has a price and color, Puzzle toy has a price and size. We have to implement class CarToy and class PuzzleToy which inherit from class Toy.
2. class ToyBox has a pointer array to store a list of toys (up to 5 items including car and puzzle) and number of items in the box.

Your task is to implement two function addItem(…) in class ToyBox. If successfully added, the function returns the current number of toys in the box. If the box is full, return -1.

For example:

Test	Result

`CarToy car(20000,red);`
`PuzzleToy puzzle(30000,small);`
`car.printType();`
`puzzle.printType();`
	`This is a car toy`
	`This is a puzzle toy`

`CarToy car(20000,red);`
`PuzzleToy puzzle(30000,small);`
 	
`ToyBox box;`
`box.addItem(car);`
`box.addItem(puzzle);`
`box.printBox();`
	`This is a car toy`
	`This is a puzzle toy`

`Toy* toy = new CarToy(30000,red);` `toy->printType();`	`This is a car toy`

Answer: (penalty regime: 0 %)

Reset answer

```
#include<iostream>
using namespace std;

enum Color
{
    red,
    green,
    blue
};
enum Size
{
    small,
    medium,
    big
};

class Toy
{
protected:
    double price;

public:
    Toy(double price)
    {
        this->price = price;
    }
```

```
    }

    virtual void printType() = 0;
    friend class ToyBox;
};

class CarToy : public Toy
{
private:
    Color color;

public:
    CarToy(double price, Color color) : Toy(price)
    {
        /*
         * STUDENT ANSWER
         */
        this->price=price;
        this->color=color;

    }

    void printType()
    {
        cout << "This is a car toy\n";
    }

    friend class ToyBox;
};

class PuzzleToy : public Toy
{
private:
    Size size;

public:
    PuzzleToy(double price, Size size) : Toy(price)
    {
        /*
         * STUDENT ANSWER
         */
        this->price=price;
        this->size=size;
    }

    void printType()
    {
        cout << "This is a puzzle toy\n";
    }

    friend class ToyBox;
};

class ToyBox
{
private:
    Toy* toyBox[5];
    int numberOfItems;

public:
    ToyBox()
    {
        /*
         * STUDENT ANSWER
         * TODO: set zero numberOfItems and nullptr toyBox
         */

         toyBox[0]=nullptr;

         numberOfItems =0;
    }

    int addItem(const CarToy& carToy)
    {
```

```
    {
        /*
         * STUDENT ANSWER
         * TODO: function add a new Car toy to the box.
                 If successfully added, the function returns the current number of toys in the box.
                 If the box is full, return -1.
         */

         if (numberOfItems==5) return -1;
         else {
             toyBox[numberOfItems]= new CarToy(carToy);
             numberOfItems++;
             return numberOfItems;
         }


    }

    int addItem(const PuzzleToy& puzzleToy)
    {
        /*
         * STUDENT ANSWER
         * TODO: function add a new Puzzle toy to the box.
                 If successfully added, the function returns the current number of toys in the box.
                 If the box is full, return -1.
         */

         if (numberOfItems==5) return -1;
         else{
            toyBox[numberOfItems]= new PuzzleToy(puzzleToy);
             numberOfItems++;
            return numberOfItems;
         }


    }

    void printBox()
    {
        for (int i = 0; i < numberOfItems; i++)
            toyBox[i]->printType();
    }
};
```

	Test	Expected	Got	

✓	CarToy car(20000,red);
	PuzzleToy puzzle(30000,small);
	car.printType();
	puzzle.printType();
		This is a car toy
		This is a puzzle toy
			This is a car toy
			This is a puzzle toy
				✓

✓	CarToy car(20000,red);
	PuzzleToy puzzle(30000,small);
	
	ToyBox box;
	box.addItem(car);
	box.addItem(puzzle);
	box.printBox();
		This is a car toy
		This is a puzzle toy
			This is a car toy
			This is a puzzle toy
				✓

✓	Toy* toy = new CarToy(30000,red); toy->printType();	This is a car toy	This is a car toy	✓

Passed all tests! ✓

Đúng

Marks for this submission: 1,00/1,00.